

3D printing with Spirulina paste workshop

4th November 2026

Agenda:

1. How to access the *3D paste Extruder app* and *3D printing spirulina* YouTube channel.
2. 3D printer unboxing.
3. 3D Printer set-up and levelling.
4. Health & Safety.
5. 3D paste extruder app.
6. Preparation of the printing paste.
7. Setting up and running a 3D printing build.
8. Flow calibration.
9. Cleaning the 3D printer.
10. 3D printer maintenance and repairs.



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3D paste Extruder app:

https://3dp-anything.github.io/Spirulina3Dprinting/

The 3D Paste Extruder App is a one-stop solution for 3D printing with spirulina paste.
It enables users to design custom cup 3D models and generate the G-code files required to operate the 3D printer.
The app also provides guidance on preparing spirulina paste, along with links to a YouTube channel, which contains step-by-step tutorials and how-to videos covering the entire 3D printing process.

3D printing spirulina YouTube channel:

https://www.youtube.com/channel/UC3OZzK21wwyKdjhjy2GVPdw

The 3D printing spirulina YouTube channel contains step-by-step tutorials and how-to videos covering the entire 3D printing process.



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3D printer unboxing: https://youtu.be/Tlk376d8lqU

3D Printer set-up and levelling: https://youtu.be/ohwO-q1kAGg

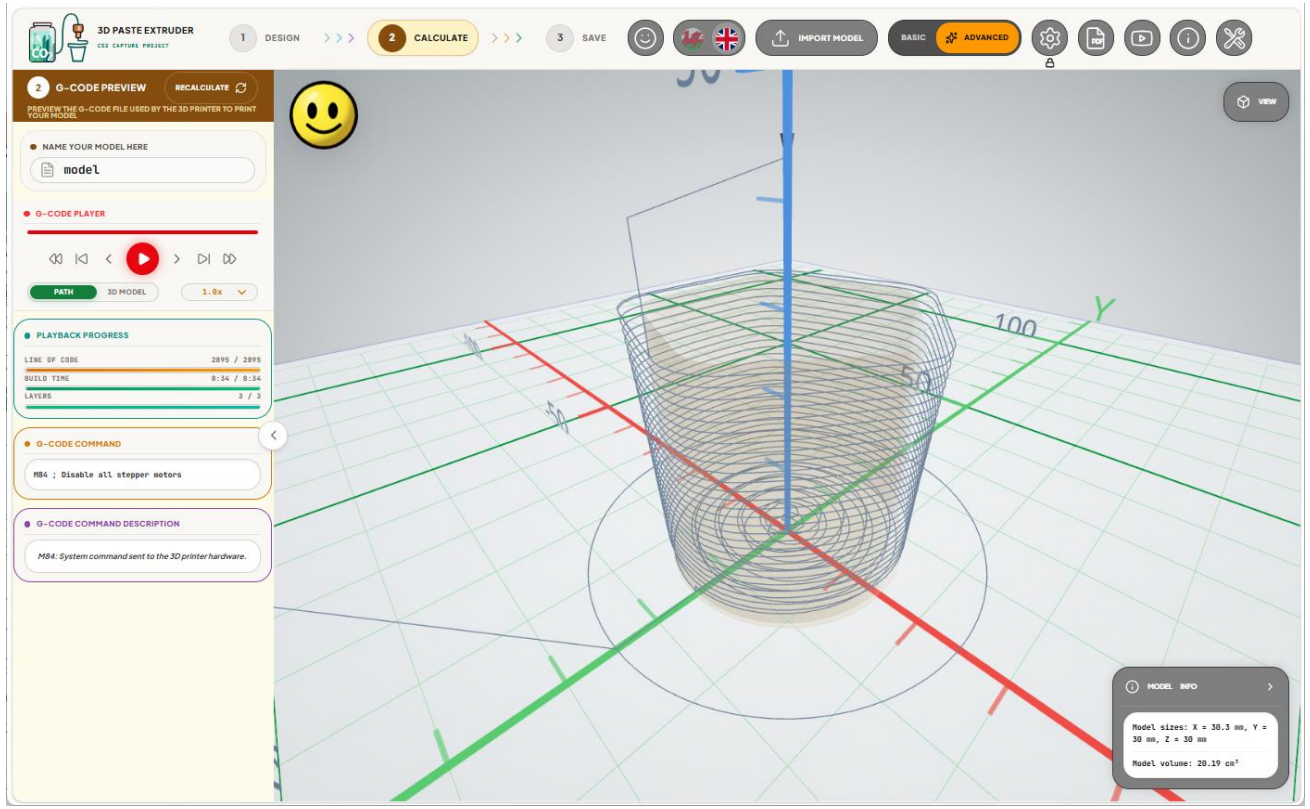
Health & Safety:

- Before operating the clay 3D printer, ensure you have read and understood all operating instructions. Always follow the manufacturer's manual instructions.
- Follow local health and safety requirements.
- Place the printer on a stable, level surface
- The printer has moving parts. When in operation, keep hands, hair, loose clothing, and jewellery away from moving parts while the printer is in operation.
- Do not touch electrical components, wiring, or connectors while the printer is powered on.
- Disconnect the power supply before carrying out maintenance, cleaning, or changing components.
- If any unusual noises, excessive vibration, material leaks, or faults are observed, stop the printer immediately and investigate the cause before continuing.
- Children and untrained users should only operate the printer under appropriate supervision.

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3D paste extruder app (1/4): https://3dp-anything.github.io/Spirulina3Dprinting/



- This app is provided free of charge and requires **no registration**.
- The G-code files generated by this app are specifically designed for use with the **Tronxy Moore 2 Clay 3D Printer**. Use with other 3D printer models or brands is entirely at the user's own risk.
- The developers accept **no liability** for any loss, damage, injury, or other consequences arising from the use of the 3D models or G-code files generated by this app.
- **Cookies are optional**, and you can continue to use the app even if you choose not to accept them.
- If you accept cookies, the app will remember your previous settings, making it quicker to use on future visits.
- Cookies are **not used for advertising, marketing, or for tracking** your activities across websites.

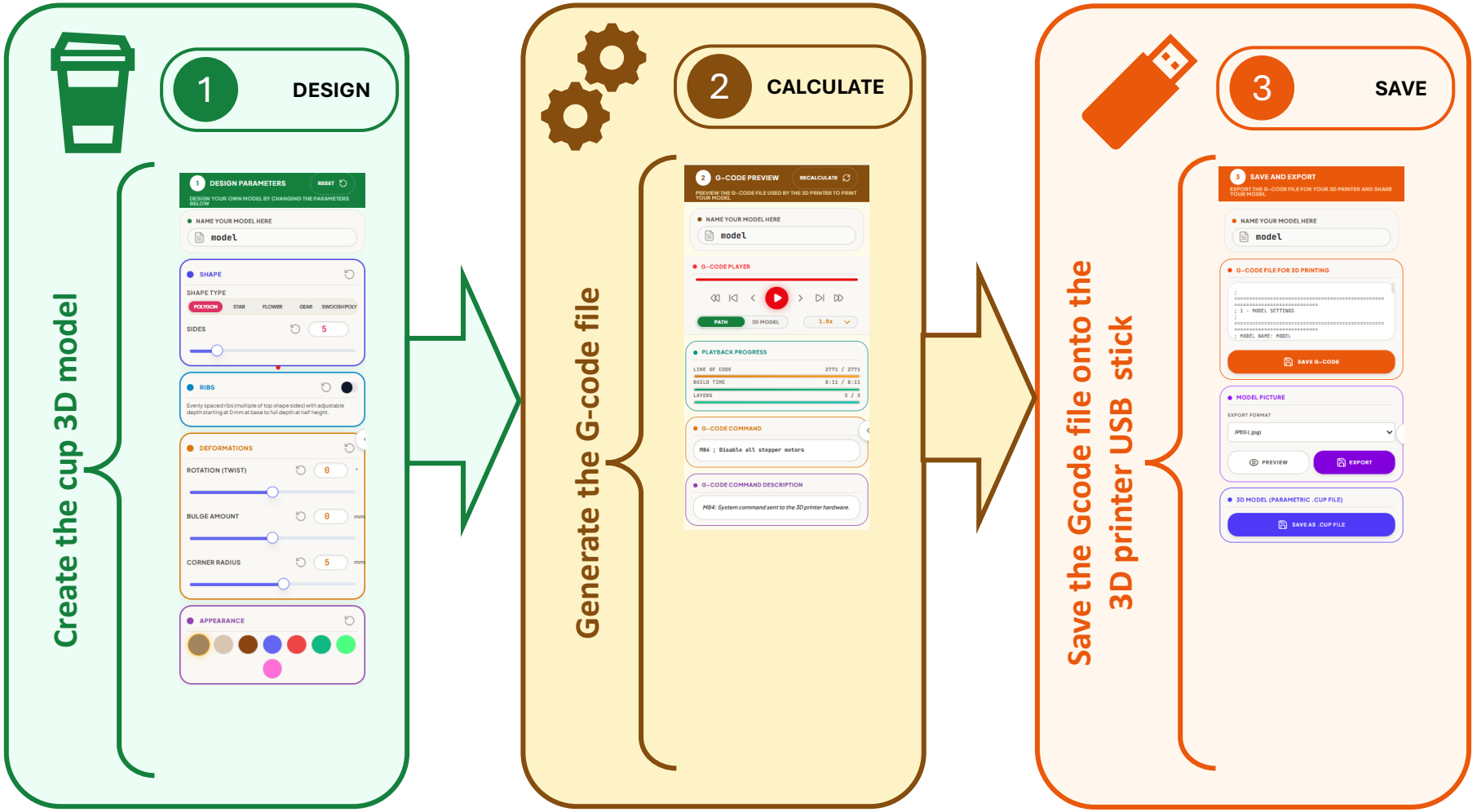
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3D paste extruder app (2/4): https://3dp-anything.github.io/Spirulina3Dprinting/





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3D paste extruder app (3/4): https://3dp-anything.github.io/Spirulina3Dprinting/

Advanced mode 

and settings menu 

Access to Advanced mode and settings menu are restricted to schools participating in the 3D Printing Spirulina Project.

Advanced Mode provides greater control over 3D printing parameters, access to export settings, and additional resources, including paste formulation guidelines, detailed procedures, instructional videos, and other supporting documentation.



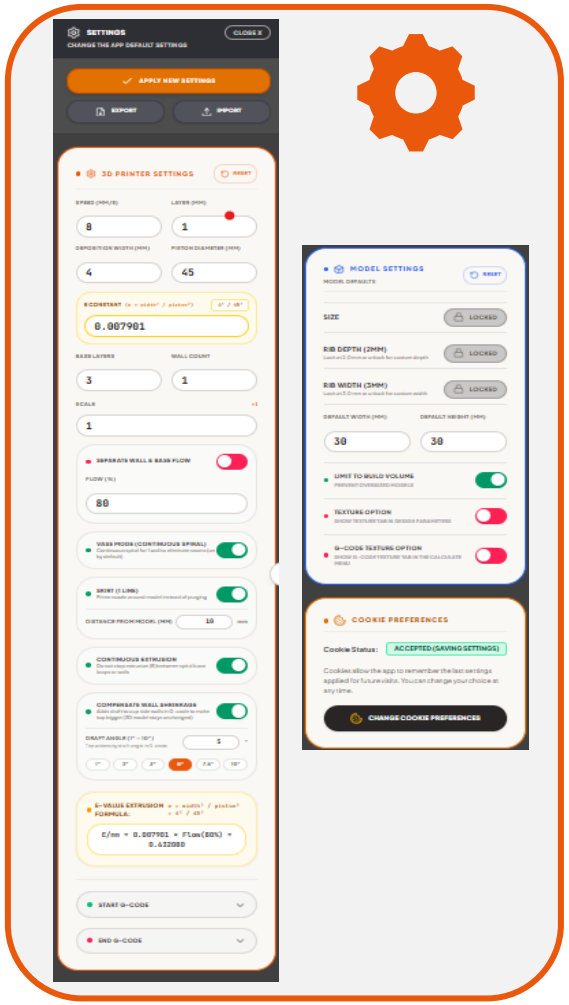
The four-digit access code is the current time shown on your device, with the colon removed. For example, if the time is 12:34, enter 1234. Entering the code unlocks access to the settings for a duration of 10 minutes.

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3D paste extruder app (4/4): https://3dp-anything.github.io/Spirulina3Dprinting/

- The G-code file used to print the cup is generated using the parameters defined in the Settings menu: BASIC ADVANCED
- Default printing parameters are provided but, due to variations in paste consistency, some adjustments will likely be required to improve print quality.
- The app includes a flow test, to help identify the optimum printing parameters for your paste formulation.
- The Settings menu also allows advanced users to override certain design restrictions, such as cup size and texture options.
- **Make sure you click the "Apply new Settings" button to validate your changes.**
- Access to the Settings menu is protected by an access code.



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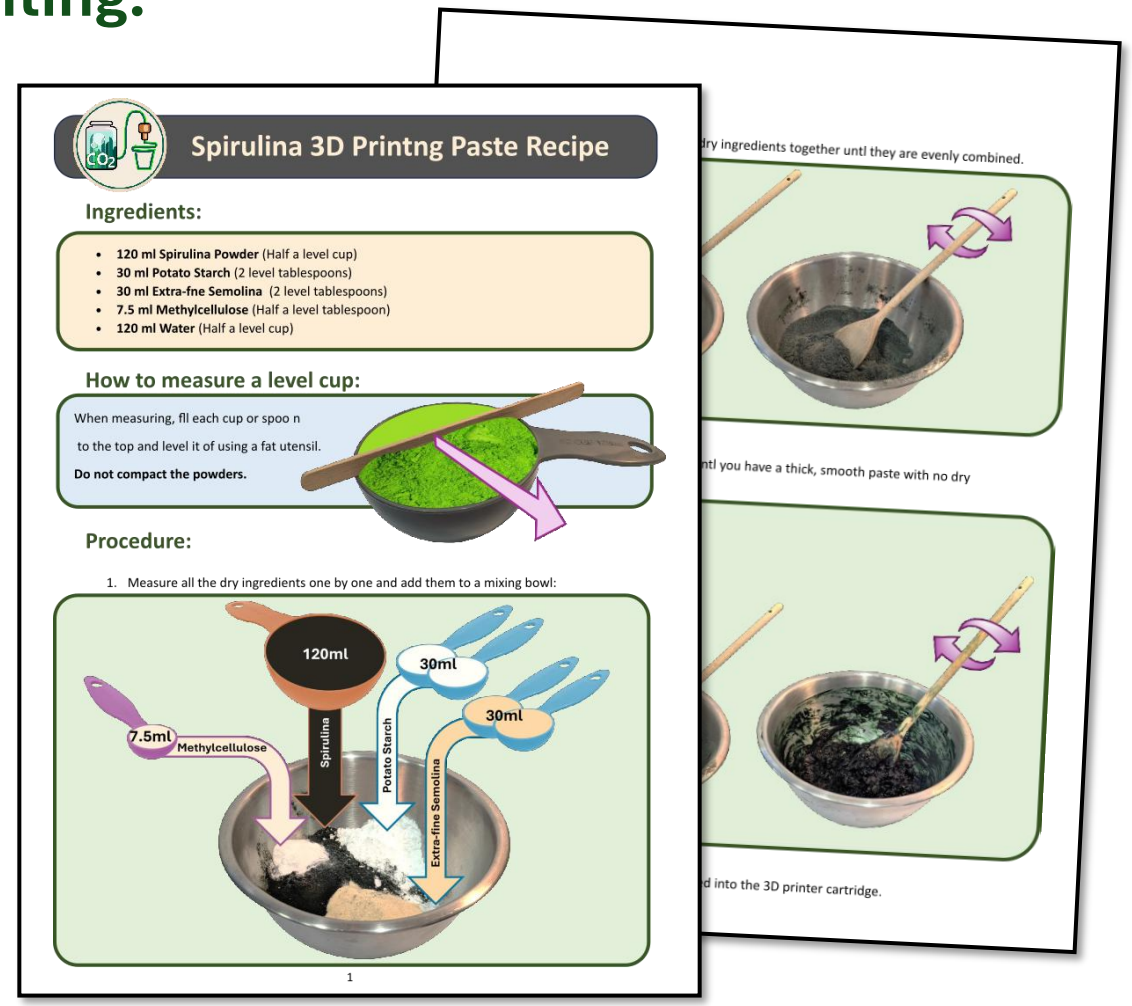


Preparing the spirulina paste for 3D printing:

- Ingredients list and preparation method are available to download directly from the app:

BASIC ADVANCED → PDF → Recipes → **Spirulina paste.pdf**

- New/improved recipes may be added as the project evolves.





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Setting up and running a 3D printing build (1/2):

1- Design your cup in the app:

2- Generate and visualise the G-code in the app:

3- Save the G-code file onto a USB memory stick:



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Setting up and running a 3D printing build (2/2):

Setting up the printer:

1. Fill the barrel with spirulina paste.
2. Mount the barrel onto the drive assembly and lock it in place by turning it ¼ turn clockwise.
3. Connect the pneumatic fittings and tube.
4. Prime the extruder: **Tools → Materials → Select "Extruder Only" mode → Extrude** until the paste reaches the print head through the tube. As the paste moves towards the print head, **gradually reduce the extrusion speed (from 10 down to 1 or 2) to avoid excessive load on the motor**.
5. Prime the nozzle: **Tools → Materials → Select "Extruder + Nozzle" mode → Extrude** until a smooth, continuous flow of paste exits the nozzle. Use a speed setting of **1**.

Start the print:

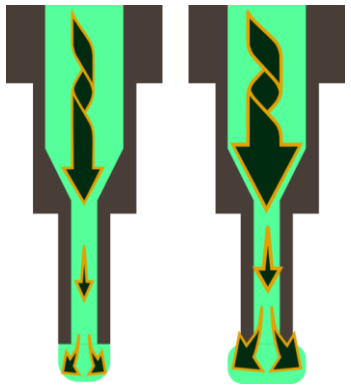
1. Insert the TF card or USB memory stick containing the G-code file(s) into the printer.
2. On the touchscreen, from the main menu, select: **Print → Select File → Confirm → Start Printing**.
3. Monitor the first one or two layers to ensure the correct amount of paste is being extruded.
4. If the cup is printing correctly, allow the print to continue. If over-extrusion or under-extrusion is observed, stop the print and perform a flow test to optimise the printing parameters.



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Flow calibration:



A flow rate that is too low will result in under-extrusion. The deposited line will be narrower than intended, which will lead to gaps, weak base/walls, poor layer bonding and unstable geometry.

A flow rate that is too high will result in over-extrusion. The deposited line will be wider than intended, causing material build-up, loss of detail, deformation of the print, and poor surface quality.



Details on how to adjust the flow parameter values using the app Flow calibration test are available here:





Flow Test → Flow calibration test.pdf

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Cleaning the 3D printer:

- Soak the printhead/nozzle in water after every print to prevent spirulina drying out and clogging the print head.
- Thoroughly clean the printer before any extended period of inactivity.
- Spirulina paste **does not stain** and can be cleaned using only water and brushes.
- Use a rod to remove as much paste as possible from the feeding tube. Then use a thin bottle brush to scrub the inside of the tube and flush it with running water to remove any remaining paste.
- Dispose of waste spirulina paste in the compost bin.
- Use brushes to clean the barrel and plunger tip.
- Cleaning the print head requires disassembly. Refer to the video linked below for instructions.

Printhead assembly/disassembly: https://youtu.be/s6HI5Q1jzl0



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3D printer maintenance and repairs:

BASIC ADVANCED

- Soak the printhead/nozzle in water after every print to prevent spirulina drying out and clogging the print head.
- Keep the 3D printer clean, especially the moving parts and guide rails.
- Thoroughly clean the printer before any extended period of inactivity.
- Regularly check the build platform levelling and adjust as required.
- Regularly check belt tension regularly and adjust if necessary.
- Always **follow the manufacturer's operating instructions** and safety guidance.
- **Technical support for the Tronxy Moore 2 Clay 3D Printer is not provided through this project/app.**
- **Any faults or technical issues should be reported directly to *support@tronxy.com*. Within your email, include the printer serial number and a detailed description of the problem.**

App issues:

- Can be reported within the app by using the feedback form accessible through the info icon ⓘ